Lecture 14: Cognitive models of semantic representation

PSYC 51.17: Models of language and communication

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Learning objectives

By the end of this lecture, you will

1. Use computational models as tools for understanding human semantic representation
2. Identify what neural-computational alignment reveals about the nature of meaning
3. Critically examine the boundaries of human understanding through a computational lens
4. Reflect on whether there are concepts humans fundamentally cannot grasp

Today's lecture

A different kind of question

Most AI discussions ask: *Can machines think like us?*

Today we flip the question: **What do machines reveal about how *we* think?**

1. Computational models as cognitive mirrors
2. The surprising alignment between brains and algorithms
3. What this tells us about the architecture of human meaning
4. The boundaries of human understanding

The mirror argument

Models as scientific instruments

When a simple algorithm (Word2Vec, BERT) captures aspects of human behavior or brain activity, this tells us something profound:

The pattern being captured was already there in human cognition.

Models don't create structure—they *reveal* structure that exists in language and thought.

The implication

If statistical co-occurrence predicts how humans judge similarity, perhaps human similarity judgments are *themselves* largely statistical.

We may be less "deep" than we imagine.

What distributional models reveal about us

Recall from Lectures 10-11

Word2Vec learns meaning purely from co-occurrence patterns in text. See Lecture 10 and Lecture 11.

The uncomfortable finding

Word2Vec predicts human similarity judgments with $r \approx 0.7$ on standard benchmarks.

This means: **~50% of variance in human semantic judgments can be explained by word co-occurrence alone.**

What does this say about the nature of human meaning?

Two interpretations

Optimistic: Language encodes deep world knowledge; models extract it.
Unsettling: Human "understanding" may be shallower than we think—pattern matching dressed up as insight.

The grounding question, inverted

The standard framing

Symbol grounding problem: How can symbols acquire meaning without sensory experience?

We usually ask this about machines. But consider:

The inverted question

How much of YOUR semantic knowledge is actually grounded?

You've never touched a quark. Never experienced the Cretaceous period. Never visited Alpha Centauri. Yet you have "concepts" of these things.

Your knowledge of most concepts is *linguistically mediated*—learned from text, speech, and symbols, not direct experience.

The implication

Perhaps humans and LLMs differ less in *kind* than in *degree*. We're both symbol manipulators—we just have more sensors.

Embodied cognition: how much does the body matter?

The embodied cognition claim

Meaning arises from bodily experience. Reading "kick" activates motor cortex. Understanding requires simulation.

But consider

- Congenitally blind individuals understand "red" and "see" semantically (if not experientially)
- People born without limbs understand "grasp" and "kick"
- Abstract concepts (justice, infinity, democracy) have no sensorimotor referent

Embodiment may be *one route* to meaning, not *the* route.

The deeper question

What is the *minimum* grounding required for genuine understanding? Is there a threshold—or is meaning a continuum?

💭 Discussion: The spectrum of grounding

Rank these from most to least grounded (5 min)

1. Your concept of "coffee" (you drink it daily)
2. Your concept of "durian" (you've read about it, maybe seen pictures)
3. Your concept of "quark" (purely theoretical, never directly observed)
4. Your concept of "justice" (abstract, no physical referent)
5. GPT-4's concept of "coffee" (learned from text)

Questions:

- Where do you draw the line between "real" understanding and symbol manipulation?
- Is your understanding of "quark" fundamentally different from GPT-4's understanding of "coffee"?
- What would it take for YOU to truly understand a quark?

Neural evidence: the alignment puzzle

Mitchell et al. (2008): Predicting brain activity from text statistics

Trained model to predict fMRI patterns from word co-occurrence with 25 sensory-motor verbs. Result: **77% accuracy** on held-out words.

Huth et al. (2016): Mapping all of semantic space

Built encoding models from word embeddings to predict brain activity during natural story listening. Found ~200 distinct semantic regions tiling the entire cortex.

The puzzle

Why should *text statistics* predict *brain activity*?

Text has no sensory information. Yet it predicts neural patterns that supposedly require embodied grounding. Either:

1. Embodied grounding leaks into language statistics, or
2. The brain's "grounded" representations are more statistical than we thought

Mitchell et al. (2008)

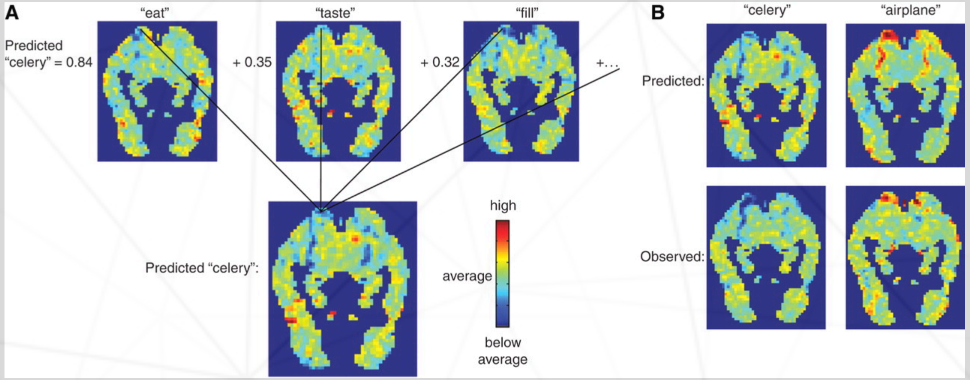
What this means for human cognition

The brain organizes semantic information in ways that parallel statistical structure in language.

This isn't evidence that the brain IS a statistical model—but it suggests the brain's solution and statistical solutions share deep structural properties.

Further reading

Mitchell et al. (2008, *Science*)



Huth et al. (2016)

The semantic atlas

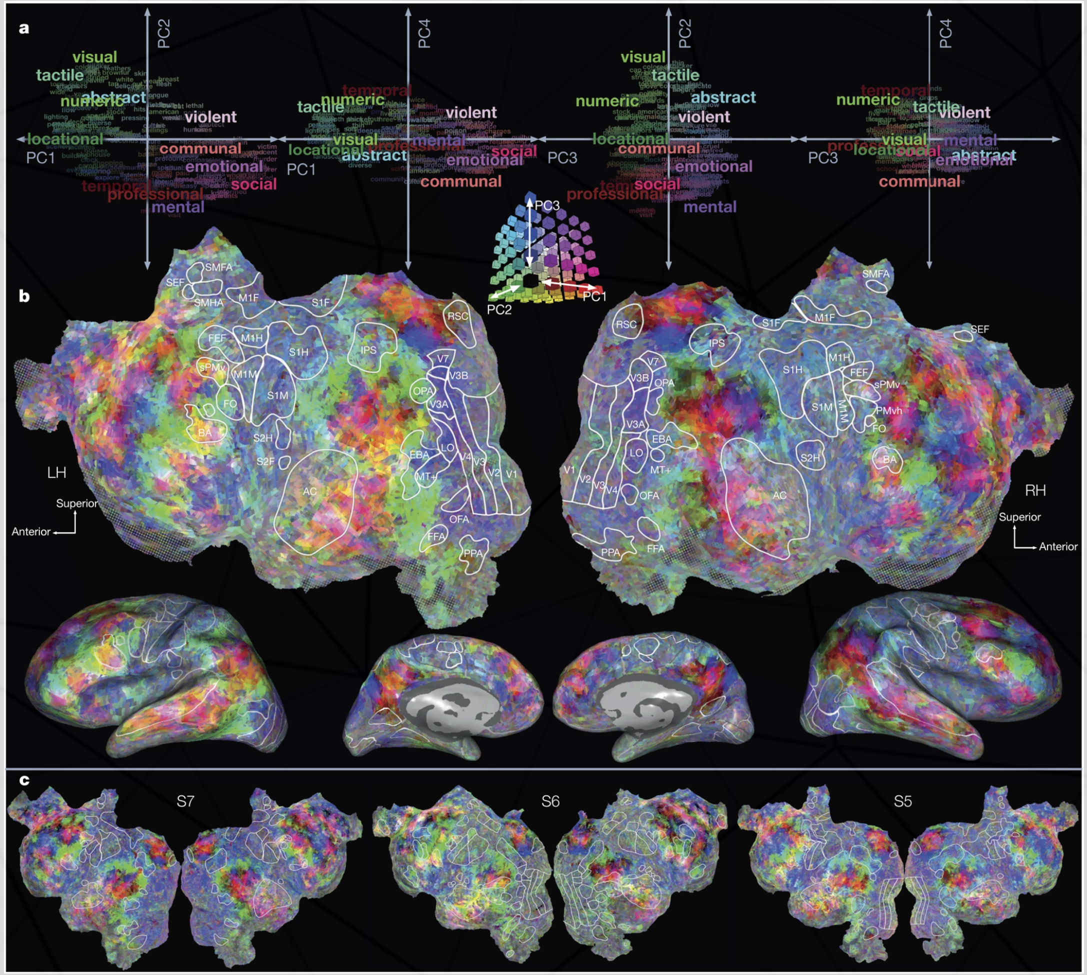
Meaning is distributed across the *entire* cortex—not localized to "language areas."

Every patch of cortex encodes semantic categories: people, places, numbers, social concepts, visual properties...

The implication

Human semantic representation is massively parallel and distributed—strikingly similar to how neural networks represent meaning.

Perhaps we invented neural networks not as alien intelligences, but as mirrors of our own cognitive architecture.



💭 Discussion: What models teach us about ourselves

Group discussion (5 min)

The Mitchell and Huth findings show that text statistics predict brain activity.

Consider:

1. If someone had never seen these results, they might assume "embodied meaning" requires fundamentally different representations than statistical learning produces. The data suggests otherwise. What does this tell us about the nature of human understanding?
2. We tend to feel that our understanding is "deep" while models are "shallow." But what if the brain's depth comes from *scale and integration* rather than a fundamentally different algorithm?
3. Could studying language models help us identify which aspects of

The limits of human understanding

A thought experiment

We easily criticize LLMs for lacking "true" understanding. But what are the limits of *human* understanding?

Candidates for the unknowable

- **Quantum superposition:** We can do the math, but can we *understand* it?
- **Four-dimensional space:** We can project it, but can we *visualize* it?
- **Exponential growth:** We systematically underestimate it (COVID-19 predictions)
- **Deep time:** 4.5 billion years is a number, not a felt quantity
- **Consciousness itself:** We experience it but cannot explain it

The pattern

Humans seem to have a "cognitive horizon"—concepts we can manipulate symbolically but cannot truly grasp.

Are humans "stochastic parrots" too?

The Bender et al. critique, redirected

LLMs are "stochastic parrots"—producing fluent language without understanding. But consider human behavior:

Evidence for human parrot-like behavior

- **Confabulation:** Patients with split brains invent plausible explanations for behaviors they don't understand
- **Motivated reasoning:** We generate arguments for conclusions we've already reached
- **Expertise illusion:** We think we understand things (zippers, toilets, politics) until asked to explain them
- **Social learning:** Much of what we "know" is repeated from others, never verified

The question

What percentage of your beliefs and utterances reflect genuine understanding versus sophisticated pattern matching and repetition?

The illusion of explanatory depth

Rozenblit & Keil (2002)

People dramatically overestimate their understanding of how everyday objects work.

The experiment: Rate your understanding of a bicycle (1-7). Now explain how it works. Re-rate your understanding.

Result: Ratings drop substantially after attempted explanation.

Implications

- We confuse *familiarity* with *understanding*
- We confuse *ability to use* with *ability to explain*
- We confuse *recognition* with *knowledge*

These are exactly the "failures" we attribute to LLMs.

Concepts humans cannot form

The cognitive closure hypothesis (McGinn, 1989)

Just as dogs cannot understand calculus (not because they're stupid, but because they lack the cognitive architecture), humans may be *constitutionally incapable* of understanding certain truths.

Possible examples

- The hard problem of consciousness (why does subjective experience exist?)
- The nature of time (why does it flow?)
- Quantum measurement (what constitutes an "observer"?)
- Why there is something rather than nothing

The uncomfortable possibility

These may not be "hard problems" awaiting clever solutions. They may be *cognitive illusions*—questions that feel meaningful but lie outside our representational capacity.

We may be LLMs hallucinating that these questions have answers.

Discussion: The boundaries of human understanding

Choose one question (7 min)

1. **Personal limits:** Is there a concept you've tried to understand but suspect you fundamentally *cannot*? What makes it feel unreachable—complexity, or something deeper?
2. **Species limits:** If humans have a "cognitive horizon," how would we know? Can a system recognize its own limitations, or is that itself beyond the limitation?
3. **The comparison:** We use "understanding" as a binary (LLMs don't, humans do). But what if understanding is a *spectrum*—and humans are simply further along than current models, not categorically different? What evidence would change your mind?
4. **The pragmatic view:** Does it *matter* whether understanding is

The role of language in thought

The Whorfian question, revisited

Does language *shape* thought, or merely *express* it?

Evidence from language models

If Word2Vec captures semantic structure from text alone, this suggests language *encodes* thought structure—it's not a neutral medium.

Implication: The structure of your language may constrain the thoughts you can think.

Languages you don't speak

There are concepts in other languages without English equivalents (saudade, hygge, Schadenfreude, wabi-sabi).

Can you *truly understand* these concepts? Or only approximate them through translation?

Are there concepts in no human language—thoughts no human has ever thought?

Meaning as a construction

The constructivist view

Meaning isn't discovered—it's *constructed* through interaction between an agent and its environment.

Both humans and LLMs construct meaning from their training experience. The difference is:

- Humans: embodied, social, extended in time, multimodal
- LLMs: disembodied, text-only, compressed, no continuous identity

The question of necessity

Which of these differences is *necessary* for "real" understanding?

If we gave an LLM a body, persistent memory, and social interaction—at what point would it cross the threshold?

Or is there no threshold—just increasingly sophisticated constructions of meaning, human and machine alike?

Summary: The mirror and the horizon

What we explored today

1. **The mirror:** Language models reveal structure in human cognition we didn't know was there
2. **The alignment:** Brain-model correlations suggest shared computational principles
3. **The inversion:** Many critiques of LLM understanding apply to humans too
4. **The horizon:** Humans have cognitive limits—concepts we can name but not truly grasp
5. **The spectrum:** "Understanding" may be continuous, not binary

The takeaway

Studying how machines fail to understand helps us see how *we* might fail to understand—and what understanding even means.

💭 Final reflection: Know thyself

Individual reflection (3 min)

The philosopher's injunction was "know thyself."

After today's lecture:

- What do you think you *truly* understand (not just recognize or use)?
- What have you realized you understand less than you thought?
- Are there important concepts you suspect you *cannot* understand—and how do you relate to that boundary?

Write a few sentences. These questions have no answers—only honest self-examination.

Questions?

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Next lecture: Attention mechanisms and the transformer revolution